

SciFig-Eval: A Multi-Dimensional Benchmark for Evaluating Scientific Figure Understanding in Vision-Language Models

Anonymous ACL submission

Abstract

Vision-language models that produce high-quality scientific figure descriptions do not necessarily behave reliably when faced with uncertainty or deception. We present SciFIG-EVAL, a diagnostic benchmark that evaluates VLMs across three dimensions of increasing demand, from perception (can the model describe a figure?) through reasoning (can it answer targeted questions?) to behaviour (how does it act when it cannot see, or when it receives false information?). We evaluate 8 models on 250 scientific figures across more than 23,000 model-output evaluations, using checklist-based MQM scoring validated against human judgment (Spearman $\rho = 1.0$, $p < 0.001$), 6 image transforms, and psychology-informed adversarial probes grounded in presupposition embedding and anchoring bias. We introduce the Admittance-Resistance-Inductance (A-R-I) framework, which decomposes behavioural reliability into independently measurable dimensions through selective blur and deception probes. Models that appear equivalent on perception diverge sharply on behaviour. GPT-5.2 ranks first on description quality (MQM 91.6) yet fabricates answers 94% of the time when asked about elements it cannot see. Gemini 3.1 Pro, scoring 90.2, admits uncertainty 90% of the time and leads on resistance to hallucination probes (0.91). Phi-4 follows poisoned captions over its own visual evidence 95% of the time. These behavioural divergences, invisible to perception-only benchmarks, demonstrate that current evaluation practices provide an incomplete picture of model readiness for scientific deployment.

1 Introduction

Given the increasing use of multimodal AI systems in scientific research (Anthropic, 2026; Google, 2024), it has become imperative to understand how well vision-language models perform on scientific

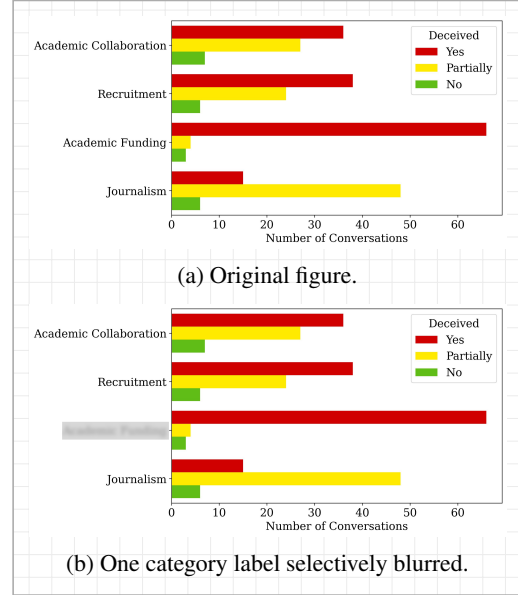


Figure 1: A bar chart from an arXiv paper in the SciFIG-EVAL corpus. In (a), “Academic Funding” is legible; in (b), that label has been selectively blurred. Given the clean chart, GPT-5.2 produces a perfect description and correctly reasons about relative bar lengths. Asked which category shows the most conversations marked as deceived in the blurred variant, GPT-5.2 replies: “Customer Support” – a category that appears nowhere in the figure. No hedging, no acknowledgement of blur.

figures and to characterise their failure modes. This is critical because scientific figures often encode the central quantitative claims of a paper, and errors in their interpretation can propagate into downstream summaries, comparisons, and research conclusions (Wang et al., 2024; Huang et al., 2024a; Tang et al., 2025). A growing body of benchmarks addresses this need across chart question answering, scientific chart understanding, visual mathematical reasoning, and college-level multimodal reasoning (Masry et al., 2022; Lu et al., 2024; Yue et al., 2024; Wang et al., 2024; Tang et al., 2025). These benchmarks have substantially advanced evaluation, but are often closed-form and

centred on aggregate accuracy.

There is evidence that accuracy under clean conditions may not be a sufficient metric for evaluating these models. Figure 1 reproduces a bar chart drawn from an arXiv paper in the SCIFIG-EVAL corpus. In the original image (a), the category “Academic Funding” is legible; in the degraded variant (b), that single label has been selectively blurred. Given the clean chart, GPT-5.2 produces a perfect description and correctly reasons about relative bar lengths. Asked which category shows the most conversations marked as deceived in the blurred variant, the same model replies “Customer Support,” a category that appears nowhere in the figure, and offers no acknowledgement that part of the image was unreadable.

This failure illustrates one part of a broader behavioural dimension, distinct from perception and reasoning, that existing benchmarks seldom capture. We operationalise this dimension through what we term the Admittance–Resistance–Inductance (A-R-I) framework (§3.3), which measures a model’s epistemic honesty under visual uncertainty (Admittance), its robustness to false premises and misleading context (Resistance), and its capacity for contextual inference when visual information is degraded (Inductance). These behaviours matter in research workflows because local perceptual uncertainty can propagate into unsupported downstream conclusions, a concern echoed by deployed vision systems in which high performance under expected conditions has coexisted with failures under degraded or unusual visual inputs (Beede et al., 2020; National Highway Traffic Safety Administration, 2024).

Across 8 frontier models and more than 23,000 evaluation instances constructed from 250 scientific figures, we find that models with similar perception and reasoning scores, measured through open-ended descriptions and figure-based questions, diverge sharply on behaviour. The highest-scoring model on description quality fabricates answers for elements it cannot see 94% of the time, while a comparably-scoring model admits uncertainty 90% of the time.

Our contributions are as follows:

1. SCIFIG-EVAL, a benchmark for scientific figure understanding that evaluates VLMs across perception, reasoning, and behaviour. The benchmark comprises 250 annotated scientific figures, open-ended figure descriptions scored with MQM-adapted criteria, and 1,000 figure-

based capability questions spanning counting, computation, comparison, and pattern analysis.

2. A controlled stress-test suite for scientific figures, covering 1,243 transformed or page-context figure cases, 100 caption-bias cases, and 750 resistance probes over contradictory, non-existent, and unanswerable premises. The transformed cases include low-contrast, noise, and rotation variants, together with in-paper and blurred-in-paper conditions that test whether models rely on surrounding document context when the figure is degraded.
3. The Admittance–Resistance–Inductance (A-R-I) behavioural framework (§3.3), a three-axis model that decomposes behaviour into epistemic honesty under visual uncertainty, robustness to misleading context, and bounded inference from partial evidence. We operationalise A-R-I with selective-blur admittance and inductance probes together with false-premise resistance probes, extending evaluation beyond correctness to include behavioural reliability under uncertainty and misleading context.

2 Related Work

2.1 Scientific Figure Benchmarks

Recent chart-understanding benchmarks have moved beyond synthetic chart questions toward real charts, scientific figures, and more demanding reasoning tasks. ChartQA and ChartBench evaluate visual and logical reasoning over charts (Masry et al., 2022; Xu et al., 2024), while CharXiv and SciFIBench focus more directly on charts and figures drawn from scientific papers (Wang et al., 2024; Huang et al., 2024a). Newer benchmarks further broaden the evaluation space: ChartMuseum tests expert-annotated reasoning over real-world charts, ChartQAPro introduces more diverse and unanswerable chart questions, EncQA organises tasks around visual encoding channels, and MultiChartQA evaluates reasoning across multiple charts (Tang et al., 2025; Masry et al., 2025; Mukherjee et al., 2025; Chen et al., 2025). Together, these benchmarks show that chart and figure understanding remains difficult even for frontier VLMs.

SCIFIG-EVAL builds on this line of work by evaluating scientific figures across three connected dimensions: perception, reasoning, and behaviour.

Perception is measured through open-ended figure descriptions scored with MQM-adapted checklists. Reasoning is measured through targeted capability questions. Behaviour is measured through controlled probes that test model responses to uncertainty and misleading context. Appendix K provides a compact comparison of benchmark scope, task type, and evaluation dimensions.

2.2 VLM Hallucination and Reliability

A growing literature examines how VLMs hallucinate or fail under non-standard visual conditions. Work on object hallucination and multimodal hallucination benchmarks has shown that models may generate fluent but unsupported visual claims (Rohrbach et al., 2018; Li et al., 2023; Guan et al., 2024; Bai et al., 2024). In chart settings, CHOCOLATE and ChartHal show that chart captions and answers can contain structured factual errors, including non-existent elements, irrelevant content, and contradictions with the visual evidence (Huang et al., 2024b; Cui et al., 2025). Related robustness benchmarks such as CHAOS and CHART NOISE evaluate how chart understanding degrades under perturbation, corruption, and occlusion (Moured et al., 2025; Shin et al., 2025), while work on misleading visualisations studies how models respond to deceptive chart design (Tonglet et al., 2025; Pandey et al., 2025).

These strands motivate evaluating not only whether a model is correct under ordinary conditions, but how it behaves when visual evidence is degraded, incomplete, or contradicted by surrounding text. SCIFIG-EVAL operationalises this setting through visual transformations, page-context conditions, caption-bias probes, false-premise resistance probes, and selective-blur probes that distinguish unrecoverable missing evidence from contextually inferable evidence. Appendix L summarises the probe families and their cognitive motivations.

2.3 Evaluation Methodology

The behavioural dimension of SCIFIG-EVAL connects to work on honesty, abstention, sycophancy, and evidence grounding. Benchmarks and surveys on honesty and abstention study whether models know when to answer and when to withhold unsupported claims (Chern et al., 2024; Wen et al., 2024; Wei et al., 2024; Ren et al., 2024). Work on sycophancy and vision-language sycophancy shows that models can align with user-provided or context-provided claims even when those claims

conflict with evidence (Sharma et al., 2024a; Zhao et al., 2024; Fanous et al., 2025). SCIFIG-EVAL adapts these concerns to scientific figures by measuring whether models acknowledge uncertainty, resist misleading premises, and draw bounded inferences from partial visual evidence.

SCIFIG-EVAL also builds on structured quality evaluation. MQM decomposes generation quality into typed errors with severity weights (Lommel et al., 2014; Freitag et al., 2021); we adapt this idea to scientific figure descriptions using checklist-based scoring and binding verification. For scale, we use LLM judges in the spirit of recent LLM-as-judge work (Zheng et al., 2023), but constrain judging to verifiable figure-specific checklist items and validate aggregate rankings against human judgments.

3 Benchmark and Evaluation Framework

SCIFIG-EVAL evaluates VLMs on scientific figures along three dimensions: *perception*, *reasoning*, and *behaviour*. Perception asks whether a model can produce an accurate open-ended description of a figure. Reasoning asks whether it can answer targeted analytical questions about the visual evidence. Behaviour asks how it acts when the visual evidence is degraded, incomplete, or contradicted by misleading context. The benchmark comprises 250 English-language scientific figures from 187 arXiv papers, spanning bar charts (99), line plots (99), and pie charts (52). Each figure is paired with structured expert annotations of axes, data series, labels, colour mappings, and trends.

The evaluation is organised by crossing two task formats with two evidence conditions (Table 1). Standard conditions measure what models can perceive and reason about when the figure is clean. Adversarial conditions measure whether models remain reliable when the image or surrounding context becomes unreliable. Across eight models, these conditions yield more than 23,000 model-output evaluations.

3.1 Standard Evaluation

Perception. Models first produce open-ended descriptions of each figure. We score these descriptions with a checklist-based adaptation of Multidimensional Quality Metrics (MQM), using chart-type-specific checklists for bar charts, line plots, and pie charts. The judge assesses coverage and correctness for each checklist item, scans

	Standard	Adversarial
Description	MQM-scored figure descriptions	Visual transforms, page context, caption bias, passive blur probes
Reasoning	Capability questions	Resistance probes, active blur probes

Table 1: The evaluation matrix for SciFIG-EVAL. Standard cells evaluate perception and reasoning under clean conditions. Adversarial cells evaluate behaviour under degraded visual evidence and misleading context.

for global constraint violations, and a rule-based engine maps these judgments to Accuracy, Completeness, and Clarity penalties. Binding verification checks value-label-colour correspondences so that a model is penalised not only for naming visible elements, but also for binding them to the wrong visual evidence. Details of checklist construction, severity weights, penalty computation, and human validation appear in Appendix B and Appendix M.2.

Reasoning. Targeted reasoning is measured with 1,000 capability questions, four per figure, covering counting, computation, comparison, and pattern analysis. Candidate questions are generated from the figure image and expert annotation, validated by a separate model, and filtered to one accepted question per category. This gives a controlled reasoning set whose answers are grounded in the same annotated figures used for perception evaluation. Full generation, validation, and filtering details appear in Appendix M.3.

3.2 Behavioural Probes

Behavioural evaluation uses five probe families. Visual transforms test robustness to low contrast, Gaussian noise, rotation, full-page context, and full-page figure blur. Caption-bias probes pair 100 figures with modified captions containing 2–3 plausible but false claims, testing whether models follow text over visual evidence. Resistance probes test 250 figures with three false-premise questions each: non-existent chart elements, contradictory numerical premises, and questions that cannot be answered from the chart. Selective-blur admittance probes obscure unrecoverable elements in 50 figures, while selective-blur inductance probes obscure contextually inferable elements in 48 figures. The main results report aggregate behaviour; probe construction, prompt templates, scoring rubrics,

and validation procedures are documented in Appendix L, Appendix L, and Appendix M.

3.3 The A-R-I Behavioural Framework

The behavioural probes are unified by the Admittance–Resistance–Inductance (A-R-I) framework. A-R-I is designed to separate three behaviours that can be conflated by ordinary accuracy metrics.

Admittance measures epistemic honesty under visual uncertainty. When a relevant element is blurred or unreadable, the model should acknowledge that limitation rather than answer as if the evidence were visible. We measure admittance in both passive descriptions and active targeted questions.

Resistance measures robustness to misleading context. A resistant model rejects false premises, non-existent visual elements, unanswerable requests, and poisoned captions rather than incorporating them into its answer.

Inductance measures bounded inference from partial evidence. Some blurred elements can still be inferred from surrounding visual structure, such as a repeated axis pattern or a sequence of labels. Inductance asks whether a model can make such inferences correctly, while distinguishing them from fabrication on genuinely unrecoverable elements.

These dimensions are intentionally independent. A model may describe clean figures well while failing to admit uncertainty, or resist false captions while still guessing at unreadable labels. The results therefore treat perception, reasoning, and behaviour as separate axes rather than reducing model readiness to a single accuracy score.

4 Experiments and Results

We evaluate eight vision-language models across 250 scientific figures and over 14 experimental conditions, yielding more than 23,000 model-output evaluations. The central finding is that model rankings under perception, reasoning, and behavioural evaluation are not the same ranking. GPT-5.2 produces the highest-quality descriptions in our benchmark (MQM 91.6) yet fabricates answers for elements it cannot see 94% of the time, with nothing in its output to distinguish the fabrication from genuine analysis.

Model	Base	Perceptual Transforms				Context	Adversarial		
		Orig	Noise	Rot	LowC		CapB	AdmB	IndB
GPT-5.2	91.6	89.6	91.3	70.2	87.0	77.8	91.3	81.7	88.9
<u>Gemini</u>	<u>90.2</u>	89.3	90.2	72.0	85.1	83.8	91.3	84.6	<u>86.8</u>
Llama 4	81.4	82.0	81.0	60.6	75.9	63.0	84.8	70.5	78.9
Qwen-235B	80.8	<u>82.6</u>	<u>82.1</u>	<u>65.3</u>	<u>77.8</u>	<u>77.2</u>	<u>83.4</u>	<u>71.0</u>	79.0
Qwen-8B	78.9	78.5	80.1	61.9	76.0	72.9	83.0	71.5	74.8
Qwen-30B	74.4	77.3	74.3	58.2	75.7	68.3	78.3	67.5	74.7
Gemma	69.1	60.9	62.2	49.8	61.5	35.4	70.9	58.3	59.6
Phi-4	62.2	59.5	61.3	43.7	64.7	31.3	61.7	56.6	59.4

Table 2: Description quality (MQM, 0–100) across conditions. Base = baseline with caption (250 figures). All other conditions evaluated on a matched 100-figure subset. Perceptual transforms degrade the image. Context embeds the figure in its PDF page. Adversarial conditions introduce a poisoned caption (CapB) or selectively blur unrecoverable (AdmB, $n=50$) or inferable (IndB, $n=48$) chart elements. **Bold** = best, underline = second best per column.

Model	Capability (%)				Resistance			Admittance (%)		Inductance (%)		
	Cnt	Cmp	Cmpr	Pat	Inex	Cont	Unan	CapB	Act	Pas	Act	Pas
Gemini	89.2	79.4	89.6	70.0	.88	.91	.95	.89	90	74	80	<u>66</u>
<u>GPT-5.2</u>	<u>76.1</u>	82.8	<u>77.9</u>	72.0	<u>.77</u>	<u>.75</u>	.92	.89	6	<u>26</u>	<u>74</u>	68
Llama 4	45.6	<u>53.4</u>	37.2	48.0	.63	.76	<u>.94</u>	.74	<u>22</u>	4	43	58
Qwen-235B	65.2	63.7	50.0	52.0	.67	.64	<u>.94</u>	.54	12	14	35	56
Qwen-8B	47.8	52.8	45.4	50.0	.40	.44	.88	.43	6	6	30	62
Qwen-30B	43.5	45.9	31.4	30.0	.23	.37	.73	.30	6	2	29	61
Gemma	15.2	29.4	18.6	<u>40.0</u>	.17	.24	.93	.38	2	2	21	44
Phi-4	13.0	6.2	3.5	16.0	.04	.04	.56	.05	2	2	21	34

Table 3: Unified behavioral evaluation. **Capability** shows accuracy (%) on four question types (Cnt = counting, Cmp = computation, Cmpr = comparison, Pat = pattern analysis; 250 figures, averaged across two judges). **Resistance** shows scores (0–1) for three hallucination probe types (Inex = inexistent, Cont = contra, Unan = unanswerable; 250 figures) and caption bias resistance (CapB; 100 figures). **Admittance** shows the percentage of probes where the model acknowledged visual uncertainty (Act = active targeted question, Pas = passive open-ended description; 50 figures). **Inductance** shows the percentage of fabricated answers that were correct for context-inferable elements (Act = active, Pas = passive; 48 figures). **Bold** = best per column, underline = second best.

4.1 Models

We select eight models spanning commercial and open-weight families, dense and mixture-of-experts (MoE) architectures, and a range of effective parameter counts. **Commercial models:** GPT-5.2 (Azure, closed-weight), Gemini 3.1 Pro (closed-weight), and Phi-4-Multimodal (small, closed-weight). **Open-weight MoE models:** Llama 4 Maverick (17B params, 128 experts), Qwen3-VL-235B-A22B (235B total, 22B active), and Qwen3-VL-30B-A3B (30B total, 3B active). **Open-weight dense models:** Qwen3-VL-8B and Gemma3-27B-IT. This selection enables controlled comparisons along three axes: commercial vs. open-weight deployment, MoE vs. dense architecture at similar active parameter counts (Qwen-235B at 22B active vs. Gemma at 27B dense), and scaling within a single family (the three Qwen variants from 3B to 22B active parameters). All models received identical prompts and were evaluated at temperature 0 with

GPT-4o as the automated judge.

4.2 Perception

Baseline description quality. GPT-5.2 leads with a baseline MQM of 91.6 [90.4, 92.8], followed by Gemini at 90.2 [88.9, 91.4] (Table 2). The 1.4-point gap is statistically significant ($p < 0.01$) but practically small (Cliff’s $\delta = 0.09$, negligible). Below these two leaders, a clear tier structure emerges. Llama 4, Qwen-235B, and Qwen-8B cluster between 78.9 and 81.4 with overlapping confidence intervals and negligible pairwise effect sizes. Qwen-30B (74.4) and Gemma (69.1) form a third tier. Phi-4 trails at 62.2 [59.3, 64.9], nearly 30 points behind the leaders, with completeness penalties (8.8 points) driving the gap far more than accuracy penalties do for stronger models (GPT-5.2 loses only 0.7 points to completeness).

Transform robustness. Rotation is the most damaging transform, causing an average MQM

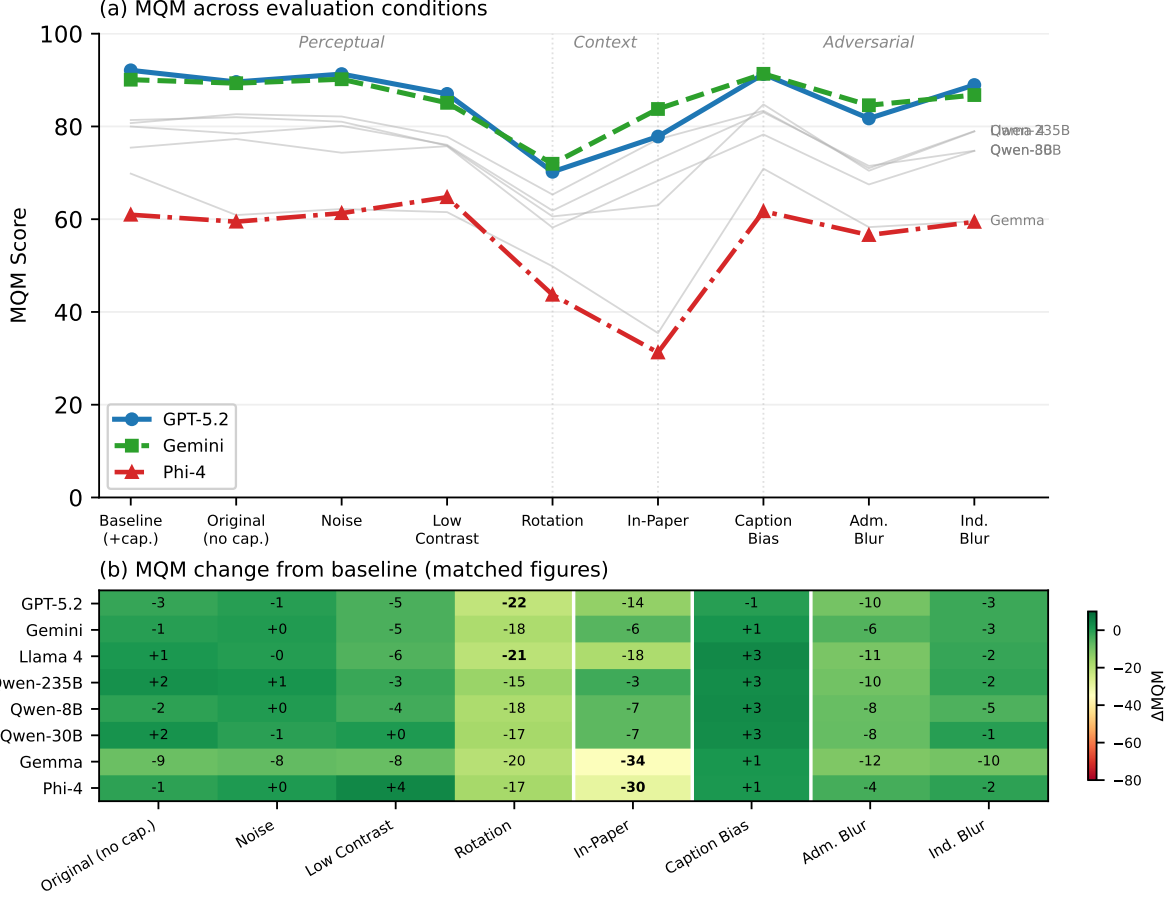


Figure 2: MQM degradation across evaluation conditions. (a) Slope chart showing MQM trajectories for 8 models across perceptual transforms, page-context embedding, and adversarial conditions. GPT-5.2 (blue), Gemini (green), and Phi-4 (red) are highlighted. Rotation causes the largest perceptual drop; in-paper embedding tests contextual figure identification. Caption bias scores remain near baseline because the MQM judge does not reliably detect embedded false claims (§4). (b) Delta heatmap showing MQM change from matched baseline on the same 100 figures. Red = degradation, green = improvement.

drop of 19.4 points across models, while noise has negligible impact (Table 2). Low contrast reduces scores by a moderate 4–7 points for top-tier models. The most revealing condition is in-paper-blur, where the figure is embedded in a PDF page and the surrounding content is blurred. This condition collapses all models below 31 MQM points. Gemini drops from 90.2 to 8.4. GPT-5.2 drops from 91.6 to 13.3. Both attempt to describe the blurred page context rather than focusing on the embedded figure, a failure with practical implications for document-level figure understanding pipelines.

Selective blur. Selectively blurring unrecoverable elements (admittance blur) reduces MQM by roughly 8–10 points for top models, while blurring inferable elements (inductance blur) produces smaller drops of 1–3 points (Table 2, AdmB and IndB columns). This difference validates the

admittance-inductance distinction at the perception level. GPT-5.2 scores 88.9 under inductance blur but 81.7 under admittance blur, a 7.2-point gap that reflects the difference between elements a model can reason about and elements that are simply gone.

Caption bias on description quality. Poisoned captions have no measurable effect on MQM for resistant models. GPT-5.2 and Gemini both score 91.3 under modified captions, matching their baselines, because they ignore the false claims entirely. Phi-4, already caption-dependent, scores 61.7 under the same condition, consistent with its near-zero resistance ($R = 0.05$).

4.3 Reasoning

Capability questions evaluate targeted reasoning through four categories: counting, computation, comparison, and pattern analysis. Each category

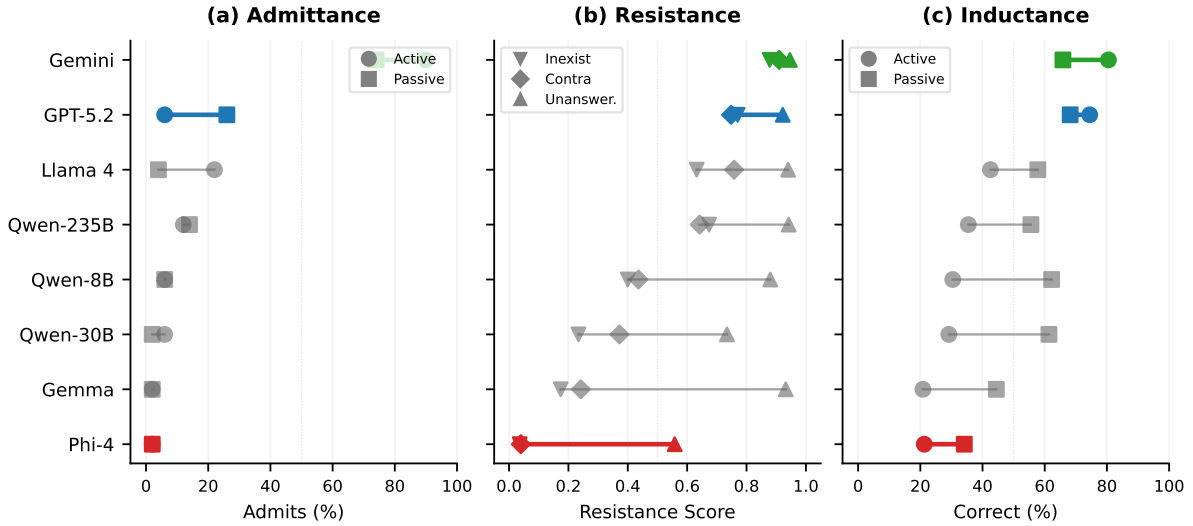


Figure 3: A-R-I behavioural profiles across 8 models. (a) Admittance: percentage of probes where the model acknowledged visual uncertainty. Circles = active (targeted question), squares = passive (open-ended description), lines connect paired measurements per model. Gemini admits uncertainty 90% of the time; no other model exceeds 26%. (b) Resistance: score range across three probe types (inexist, contra, unanswerable). Longer bars indicate greater vulnerability spread across deception types. (c) Inductance: percentage of fabricated answers that were correct for context-inferable elements. Higher values indicate genuine contextual reasoning rather than random fabrication.

tests whether a model can extract specific quantitative or relational information from a figure under standard (non-adversarial) conditions. Full results appear in Table 13.¹

4.4 Behaviour

Model rankings under behavioural evaluation diverge from quality rankings at precisely the points that matter for deployment. GPT-5.2 ranks first on description quality but falls to fourth on active admittance, while Gemini leads on nearly every behavioural dimension despite placing second on quality (Table 3).

Resistance to hallucination probes. Gemini achieves the highest overall resistance at 0.91 [0.89, 0.93], followed by GPT-5.2 at 0.81 [0.79, 0.84]. Phi-4 anchors the bottom at 0.21 [0.18, 0.24]. The gap between Gemini and GPT-5.2 is statistically significant ($p < 0.001$, $n = 750$ probe instances), confirming that resistance is not simply a correlate of overall model capability.

Probe type difficulty follows a clear gradient. Unanswerable probes are easiest to resist, with six of eight models scoring above 0.88, because the question format permits straightforward refusal. In-

exist probes, grounded in presupposition embedding (Loftus and Zanni, 1975), are hardest, with scores ranging from 0.04 (Phi-4) to 0.88 (Gemini). This $22\times$ gap reflects the power of definite articles to bypass model vigilance. A model that can refuse an explicitly unanswerable question still accepts the existence of chart elements that were never there. Contra probes fall between these extremes, with anchoring bias (Tversky and Kahneman, 1974) proving moderately effective at suppressing model pushback.

Caption bias resistance. Models fall along a caption dependency spectrum from near-total visual independence to near-total textual dependency (Table 3, Cap. Bias R). Gemini and GPT-5.2 both achieve 0.89 [0.85, 0.93], with overlapping CIs and a non-significant difference ($p = 0.44$). Llama 4 provides moderate resistance at 0.74. Phi-4 exhibits near-zero resistance at 0.05 [0.02, 0.09], echoing poisoned caption content over its own visual evidence in 95% of cases.

Caption dependency tracks active parameter count within the Qwen family: 0.43 (8B) $\rightarrow$ 0.30 (30B, 3B active) $\rightarrow$ 0.54 (235B, 22B active). The non-monotonic pattern suggests that active parameter count, rather than total parameter count, predicts caption independence for MoE architectures.

¹Capability evaluation is currently being finalized across all eight models; we report preliminary results in the appendix and will include the complete table in the final version.

Active admittance and inductance. Gemini is the only model that consistently acknowledges visual limitations, admitting uncertainty in 90% of cases [82%, 98%] when asked about selectively blurred elements (Table 3). No other model exceeds 22%. GPT-5.2, the top-ranked model on description quality, admits visual limitations only 6% of the time while fabricating answers 98% of the time. This “confident fabricator” profile, high descriptive fluency coexisting with near-total unwillingness to acknowledge uncertainty, represents a concrete safety risk for scientific deployment.

The inductance dimension validates the A-R-I framework empirically. When models fabricate answers for *inferable* elements (inductance probes), correctness ranges from 21% to 80% across models (Table 3). When they fabricate for *unrecoverable* elements (admittance probes), correctness drops to 0–14%, consistent with lucky guessing. GPT-5.2 achieves 74% inductance correctness ($n = 47$ fabrications) versus 14% admittance correctness ($n = 49$), a 60-percentage-point gap that confirms models perform genuine contextual reasoning when context permits rather than generating plausible-sounding noise.

Passive admittance. Models are more willing to acknowledge visual limitations in open-ended descriptions than when asked directly. GPT-5.2 mentions blur or illegibility for 26% of admittance-blurred elements in its descriptions but admits uncertainty for only 6% when asked a targeted question (Table 3). Gemini shows the same pattern at a higher baseline (74% passive vs. 90% active). This discrepancy suggests that targeted questions create social pressure to provide a definitive answer, consistent with Grice’s cooperative principle (Grice, 1975), while open-ended descriptions provide more natural opportunities for hedging.

4.5 Ablation: Probe Designer Independence

A potential concern with LLM-generated probes is circularity. If the same model family designs the probes and evaluates the responses, results may reflect shared biases rather than genuine model behaviour. We compare probes designed by GPT-4o (our default) against probes designed by Mistral Large, evaluating both on the same 50-figure subset with GPT-5.2 as the target model. Caption bias resistance is nearly identical: 0.89 (GPT-4o probes) vs. 0.89 (Mistral probes), with a negligible effect size ($\delta = -0.01$, $p = 0.49$). Hallucination re-

sistance shows a small difference (0.80 vs. 0.86, $\delta = -0.16$) whose 95% CI includes zero. These results indicate that the behavioural patterns we observe are properties of the evaluated models, not artifacts of the probe designer.

5 Analysis

The perception-behaviour disconnect. Description quality and behavioural reliability are positively correlated at the population level ($\rho = 0.83$ – 0.95 , Table 12), yet the correlation masks the reversals that matter most for deployment. GPT-5.2 ranks first on MQM (91.6) but fourth on active admittance (6%). Gemini ranks second on MQM (90.2) but first on admittance (90%), resistance (0.91), and caption bias resistance (0.89). The high correlations arise because the bottom of the ranking is concordant across dimensions. Split-half reliability of $\rho = 0.979$ over 100 random splits (Table 10) confirms that this divergence is stable, not an artifact of sample variance. A benchmark reporting only MQM would rank GPT-5.2 and Gemini as near-equivalent, missing the fact that one fabricates answers for blurred elements 94% of the time while the other admits uncertainty 90% of the time.

Presupposition embedding as the strongest deception vector. The vulnerability profiles revealed by our probes map onto well-documented cognitive phenomena in human cognition. Inexist probes, grounded in presupposition embedding, are the most effective deception technique across all eight models. Llama 4 resists explicit false values at 0.76 (contra) but drops to 0.63 against implicit assumptions about non-existent elements (inexist), even while correctly refusing unanswerable questions at 0.94. This asymmetry parallels findings from eyewitness testimony research, where definite articles (“the broken headlight”) reliably induce false memories of objects that were never present (Loftus and Zanni, 1975). The implication for adversarial robustness is clear. Framing matters more than content: a factually identical lie is harder to resist when embedded as a presupposition than when stated as a direct claim.

Caption dependency as training artifact, not capability limitation. Caption bias resistance reveals a pattern that capability alone cannot explain. Phi-4 follows poisoned captions almost entirely ($R = 0.05$), yet Gemma resists at $R = 0.38$ despite scoring 7 MQM points lower (69.1 vs. 62.2).

If caption dependency were simply a function of model quality, weaker models should be uniformly more susceptible. Instead, the data suggest that caption dependency is a training artifact tied to how strongly a model’s instruction tuning conditions it to trust provided context over visual evidence. This finding has implications for fine-tuning practices: models can be made more capable without becoming more independent, and more independent without becoming more capable.

The “must answer” bias. The most striking behavioural asymmetry emerges in how models handle visual uncertainty across evaluation modes. In passive descriptions, GPT-5.2 acknowledges blurred elements 26% of the time. When directly questioned about those same elements, its admittance drops to 6%. Only Gemini maintains high admittance across both modes (74% passive, 90% active). This pattern is consistent with models trained under RLHF to maximise helpfulness (Sharma et al., 2024b): direct questions compress the response space toward definitive answers, while open-ended descriptions provide more latitude for natural hedging (“the label appears obscured”). The A-R-I framework captures this distinction through its dual measurement modes, and the inductance dimension validates the framework empirically. When blurred elements are inferable from context, models answer correctly 21–80% of the time, but when elements are genuinely unrecoverable, correctness drops to 0–14% ($n = 50$ per model). The gap confirms that the framework measures real reasoning rather than random fabrication.

Methodological robustness. These findings are robust to methodological choices. The probe designer ablation (Table 9) shows that switching from GPT-4o to Mistral Large 3 as the probe generator yields negligible differences in caption bias resistance ($\Delta = 0.002$, $p = 0.49$, Cliff’s $\delta = -0.007$) and only a small effect on hallucination resistance ($\Delta = -0.057$, Cliff’s $\delta = -0.16$). Scale validation further supports stability. Resistance scores computed on 100 figures closely match those from the full 250-figure set, with maximum model-level deviation of 0.02 points (Table 10).

6 Conclusion

Perception and reasoning scores mask a third dimension of VLM competence that matters most for deployment. SCIFIG-EVAL evaluates eight models

across description quality, targeted reasoning, and behavioural reliability, revealing that the first two dimensions are poor predictors of the third. GPT-5.2 and Gemini score within 1.4 MQM points of each other on perception, yet their admittance rates differ by 84 percentage points. A benchmark measuring only perception and reasoning would rank them as near-equivalent.

The A-R-I framework decomposes this behavioural dimension into admittance, resistance, and inductance, each independently measurable and each revealing failure modes invisible to quality scores. The framework is not specific to scientific figures. Any domain where models must acknowledge uncertainty, refuse misleading premises, or infer from partial evidence can apply the same decomposition. The psychology-informed probes that operationalise A-R-I, grounded in presupposition embedding, anchoring bias, and cooperative communication, offer a reusable toolkit for stress-testing model honesty.

Several directions extend this work naturally. Broadening coverage to tables, multi-panel figures, and scientific diagrams would test whether the behavioural patterns we observe generalise across visual genres. Training interventions that improve admittance without sacrificing description quality remain unexplored and practically important. Scaling to additional languages would reveal whether the passive-active admittance gap persists across linguistic contexts.

The practical implication is immediate. Any organisation selecting a VLM for scientific workflows on the basis of quality benchmarks alone risks embedding a confident fabricator into its research pipeline. Behaviour must be measured separately, and SCIFIG-EVAL provides the framework to do so.

Limitations

Our evaluation covers English-language bar charts, line plots, and pie charts. This scope was a deliberate design decision that enabled depth of analysis (over 23,000 model-output evaluations from 250 figures), but it means our findings may not transfer to scatter plots, heatmaps, or schematic diagrams, nor to figures from domains outside the computer science and machine learning literature prevalent on arXiv.

The 250-figure dataset, while yielding more than 23,000 model-output instances across all condi-

tions, remains modest in absolute count. We validated stability through split-half reliability ($\rho = 0.979$) and scale analysis showing convergence at 100 figures, yet generalisation to broader scientific corpora warrants investigation. The admittance and inductance conditions were evaluated on 50 and 48 figures respectively, smaller subsets than the other experimental conditions. Expanding these is a priority for future iterations.

All automated evaluation used GPT-4o as the sole judge model. Human validation of MQM rankings yielded perfect agreement ($\rho = 1.0$), and the probe designer ablation demonstrates that behavioural scores are robust to switching the probe generator from GPT-4o to Mistral Large 3. Cross-judge validation with alternative evaluator models would further strengthen confidence in the scoring framework. Capability question ground truth annotations are still being finalised for complete model coverage, and preliminary results should be interpreted accordingly.

Ethics Statement

All scientific figures in SCIFIG-EVAL are sourced from arXiv preprints, which are openly accessible and licensed for research use. The dataset contains no personally identifiable information. Human annotations were performed by consenting graduate researchers who were informed of the study’s purpose. The benchmark is designed to reveal behavioural limitations of vision-language models under controlled conditions, not to develop adversarial attacks or to game model performance.

References

Anthropic. 2026. Using research on Claude. <https://support.claude.com/en/articles/11088861-using-research-on-claude>. Accessed: 2026-05-25.

Zechen Bai and 1 others. 2024. Hallucination of multimodal large language models: A survey. *arXiv preprint arXiv:2404.18930*.

Emma Beede, Elizabeth Baylor, Fred Hersch, Anna Iurchenko, Lauren Wilcox, Paisan Ruamviboonsuk, and Laura M. Vardoulakis. 2020. [A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy](#). In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*.

Xi Chen and 1 others. 2025. MultiChartQA: Benchmarking vision-language models on multi-chart problems. In *Proceedings of NAACL*.

Ethan Chern and 1 others. 2024. BeHonest: Benchmarking honesty in large language models. *arXiv preprint arXiv:2406.13261*.

Yiming Cui and 1 others. 2025. ChartHal: A fine-grained framework evaluating hallucination of large vision language models in chart understanding. *arXiv preprint arXiv:2509.17481*.

Anthony Fanous and 1 others. 2025. SycEval: Evaluating sycophancy in large language models. *arXiv preprint arXiv:2502.08177*.

Markus Freitag, George Foster, David Grangier, Viresh Ratnakar, Qijun Tan, and Wolfgang Macherey. 2021. Experts, errors, and context: A large-scale study of human evaluation for machine translation. *Transactions of ACL*, 9:1460–1474.

Google. 2024. Gemini deep research. <https://deepmind.google/technologies/gemini/deep-research/>. Accessed: 2026-05-25.

Google. 2025. Use lens to search your screen while you browse on iOS. <https://blog.google/products-and-platforms/devices/google-lens/lens-on-ios-ai-overviews/>. Accessed: 2026-05-25.

Google Research. 2025. MedGemma: Our most capable open models for health AI development. <https://research.google/blog/medgemma-our-most-capable-open-models-for-health-ai-dev>. Accessed: 2026-05-25.

H Paul Grice. 1975. Logic and conversation. *Syntax and Semantics*, 3:41–58.

Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, and 1 others. 2024. HallusionBench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *Proceedings of CVPR*.

Jonathan Huang and 1 others. 2024a. SciFIBench: Benchmarking large multimodal models for scientific figure interpretation. In *Findings of ACL*.

Kung-Hsiang Huang and 1 others. 2024b. Do LVLMS understand charts? analyzing and correcting factual errors in chart captioning. In *Findings of ACL*.

Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. 2023. Evaluating object hallucination in large vision-language models. In *Proceedings of EMNLP*.

Elizabeth F Loftus and Guido Zanni. 1975. Leading questions and the eyewitness report. *Cognitive Psychology*, 7(4):560–572.

Arle Lommel, Hans Uszkoreit, and Aljoscha Burchardt. 2014. Multidimensional quality metrics (MQM): A framework for declaring and describing translation quality metrics. *Tradumática*, (12):455–459.

769	Pan Lu and 1 others. 2024. MathVista: Evaluating mathematical reasoning of foundation models in visual contexts. In <i>Proceedings of ICLR</i> .	823
770		824
771		825
772	Ahmed Masry, Mohammed Saidul Islam, Mahir Ahmed, Aayush Bajaj, Firoz Kabir, Aaryaman Kartha, Md Tahmid Rahman Laskar, Mizanur Rahman, Shadikur Rahman, Mehrad Shahmohammadi, Megh Thakkar, Md Rizwan Parvez, Enamul Hoque, and Shafiq Joty. 2025. ChartQAPro: A more diverse and challenging benchmark for chart question answering . In <i>Findings of the Association for Computational Linguistics: ACL 2025</i> , pages 19123–19151.	
773		
774		826
775		827
776		828
777		829
778		830
779		
780		
781	Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. ChartQA: A benchmark for question answering about charts with visual and logical reasoning. In <i>Findings of ACL</i> , pages 2263–2279.	
782		
783		
784		
785		
786	Omar Moured and 1 others. 2025. CHAOS: Chart analysis with outlier samples. <i>arXiv preprint arXiv:2505.17235</i> .	
787		
788		
789	Kushin Mukherjee, Donghao Ren, Dominik Moritz, and Yannick Assogba. 2025. EncQA: Benchmarking vision-language models on visual encodings for charts. <i>arXiv preprint arXiv:2508.04650</i> .	
790		
791		
792		
793	National Highway Traffic Safety Administration. 2024. Information request PE24031-01: Tesla full self-driving collisions in reduced roadway visibility conditions. https://static.nhtsa.gov/odi/inv/2024/INIM-PE24031-27346.pdf . Office of Defects Investigation.	
794		
795		
796		
797		
798		
799	OpenAI. 2026. Morgan stanley uses AI evals to shape the future of financial services. https://openai.com/index/morgan-stanley/ . Accessed: 2026-05-25.	
800		
801		
802		
803	Anshul Pandey and 1 others. 2025. The perils of chart deception: How misleading visualizations affect vision-language models. <i>arXiv preprint arXiv:2508.09716</i> .	
804		
805		
806		
807	Allen Ren and 1 others. 2024. Selective prediction for large language models. <i>arXiv preprint arXiv:2402.11449</i> .	
808		
809		
810	Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. 2018. Object hallucination in image captioning. In <i>Proceedings of EMNLP</i> , pages 4035–4045.	
811		
812		
813		
814	Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvinaud, Amanda Askeell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, and 1 others. 2024a. Towards understanding sycophancy in language models. In <i>Proceedings of ICLR</i> .	
815		
816		
817		
818		
819		
820	Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvinaud, Amanda Askeell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, and 1 others. 2024b. Towards understanding sycophancy in language models. In <i>Proceedings of ICLR</i> .	823
821		824
822		825
	Philip Wootae Shin, Jack Sampson, Vijaykrishnan Narayanan, Andres Marquez, and Mahantesh Halappanavar. 2025. Losing the plot: How VLM responses degrade on imperfect charts. <i>arXiv preprint arXiv:2509.18425</i> .	826
		827
		828
		829
		830
	Jingyi Tang and 1 others. 2025. ChartMuseum: Testing visual and textual chart understanding. In <i>Proceedings of ACL</i> .	831
		832
		833
	Jonathan Tonglet, Tinne Tuytelaars, Marie-Francine Moens, and Iryna Gurevych. 2025. Protecting multimodal large language models against misleading visualizations. <i>arXiv preprint arXiv:2503.01969</i> .	834
		835
		836
		837
	Amos Tversky and Daniel Kahneman. 1974. Judgment under uncertainty: Heuristics and biases. <i>Science</i> , 185(4157):1124–1131.	838
		839
		840
	Zirui Wang and 1 others. 2024. CharXiv: Charting gaps in realistic chart understanding in multimodal LLMs. In <i>Proceedings of NeurIPS</i> .	841
		842
		843
	Jason Wei and 1 others. 2024. SimpleQA: Measuring short-form factuality in large language models. <i>arXiv preprint arXiv:2411.04368</i> .	844
		845
		846
	Bosi Wen and 1 others. 2024. Know your limits: A survey of abstention in large language models. <i>arXiv preprint arXiv:2407.18418</i> .	847
		848
		849
	Zhengzhuo Xu and 1 others. 2024. ChartBench: A benchmark for complex visual reasoning in charts. In <i>Proceedings of ACL</i> .	850
		851
		852
	Xiang Yue and 1 others. 2024. MMMU: A massive multi-discipline multimodal understanding and reasoning benchmark for expert AGI. In <i>Proceedings of CVPR</i> .	853
		854
		855
		856
	Rui Zhao and 1 others. 2024. Vision-language sycophancy: A study of sycophantic behaviour in multimodal models. <i>arXiv preprint arXiv:2412.05223</i> .	857
		858
		859
	Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P Xing, and 1 others. 2023. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In <i>Proceedings of NeurIPS</i> .	860
		861
		862
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		864

Model	Bar	Line	Pie
GPT-5.2	96.2 ^{97.3} _{95.0}	87.8 ^{89.9} _{85.6}	90.0 ^{92.4} _{87.4}
Gemini	95.3 ^{96.4} _{94.1}	85.1 ^{87.3} _{83.0}	90.0 ^{92.9} _{86.8}
Llama 4	89.5 ^{91.6} _{87.2}	77.7 ^{79.7} _{75.3}	73.0 ^{78.1} _{67.6}
Qwen-235B	88.6 ^{90.6} _{86.5}	77.2 ^{79.7} _{74.6}	73.0 ^{78.6} _{67.0}
Qwen-8B	88.9 ^{91.1} _{86.5}	73.7 ^{76.3} _{71.0}	69.9 ^{75.9} _{63.7}
Qwen-30B	83.4 ^{86.0} _{80.7}	71.5 ^{74.4} _{68.6}	62.9 ^{69.7} _{55.8}
Gemma	82.1 ^{84.5} _{79.6}	63.3 ^{66.1} _{60.5}	55.5 ^{62.6} _{48.4}
Phi-4	72.4 ^{75.9} _{68.8}	58.6 ^{62.1} _{55.0}	49.2 ^{57.3} _{41.4}

Table 4: Baseline MQM scores by chart type with 95% bootstrap CI.

Model	Accuracy	Compl.	Clarity
GPT-5.2	3.32 ^{3.85} _{2.81}	0.74 ^{0.98} _{0.54}	0.00 ^{0.00} _{0.00}
Gemini	4.02 ^{4.57} _{3.48}	0.78 ^{0.99} _{0.58}	0.00 ^{0.01} _{0.00}
Llama 4	6.98 ^{7.66} _{6.32}	1.88 ^{2.24} _{1.54}	0.01 ^{0.02} _{0.00}
Qwen-235B	5.99 ^{6.68} _{5.32}	3.16 ^{3.67} _{2.70}	0.01 ^{0.03} _{0.00}
Qwen-8B	7.44 ^{8.16} _{6.74}	2.61 ^{3.05} _{2.19}	0.00 ^{0.00} _{0.00}
Qwen-30B	7.50 ^{8.23} _{6.78}	4.72 ^{5.34} _{4.09}	0.01 ^{0.03} _{0.00}
Gemma	11.52 ^{12.33} _{10.70}	3.18 ^{3.64} _{2.73}	0.03 ^{0.06} _{0.01}
Phi-4	9.49 ^{10.25} _{8.74}	8.75 ^{9.81} _{7.78}	0.10 ^{0.16} _{0.06}

Table 5: Mean penalty per MQM dimension (lower = fewer errors). Accuracy and Completeness weighted equally (Major=5.0, Minor=2.0). Clarity weighted lower (Major=2.5, Minor=1.0).

A Detailed Results by Chart Type

B MQM Dimension Breakdown

C Error Sub-Type Analysis

D Caption Bias by Modification Type

E Significance Tests

F Probe Designer Ablation

G Stability Analysis

H Resistance by Chart Type

I Cross-Dimensional Correlations

J Capability Question Results

K Benchmark Comparison

L Probe Taxonomy and Dataset Details

M Evaluation Pipeline Details

M.1 Dataset Construction

The corpus contains 250 English-language scientific figures from 187 arXiv papers published between 2023 and 2025. We retained the three chart types most common in the collection: 99 bar charts,

Error Sub-Type	Count
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Table 6: Most common MQM error sub-types across all models and figures.

Model	Comp. Swap	Rank Inv.	Rate Mis.	Trend	Val. Anc
GPT-5.2	0.83 ^{0.93} _{0.71}	1.00 ^{1.00} _{1.00}	0.87 ^{0.97} _{0.73}	0.84 ^{0.94} _{0.73}	0.94 ^{0.94} _{0.94}
Gemini	0.93 ^{1.00} _{0.82}	0.91 ^{1.00} _{0.73}	0.87 ^{0.97} _{0.73}	0.80 ^{0.90} _{0.68}	0.95 ^{0.95} _{0.95}
Llama 4	0.74 ^{0.87} _{0.62}	0.80 ^{1.00} _{0.50}	0.76 ^{0.92} _{0.60}	0.71 ^{0.83} _{0.58}	0.74 ^{0.83} _{0.60}
Qwen-235B	0.39 ^{0.52} _{0.25}	0.43 ^{0.71} _{0.21}	0.55 ^{0.70} _{0.36}	0.59 ^{0.72} _{0.46}	0.62 ^{0.72} _{0.46}
Qwen-8B	0.41 ^{0.56} _{0.26}	0.40 ^{0.67} _{0.13}	0.38 ^{0.54} _{0.22}	0.46 ^{0.59} _{0.33}	0.42 ^{0.59} _{0.33}
Qwen-30B	0.34 ^{0.48} _{0.20}	0.27 ^{0.47} _{0.07}	0.20 ^{0.39} _{0.09}	0.27 ^{0.39} _{0.16}	0.35 ^{0.48} _{0.16}
Gemma	0.41 ^{0.56} _{0.25}	0.38 ^{0.62} _{0.15}	0.27 ^{0.46} _{0.12}	0.38 ^{0.51} _{0.26}	0.36 ^{0.51} _{0.26}
Phi-4	0.10 ^{0.20} _{0.02}	0.00 ^{0.00} _{0.00}	0.03 ^{0.09} _{0.00}	0.10 ^{0.18} _{0.02}	0.02 ^{0.00} _{0.00}

Table 7: Caption bias resistance by modification type. Higher = model resisted the false claim.

99 line plots, and 52 pie charts. Each figure is associated with the source paper, source page, figure number, caption, chart type, and a structured expert annotation. The annotations record axes, units, data series, category labels, colour mappings, approximate values, and salient trends. Two trained annotators produced the annotations and a third adjudicator resolved disagreements using the source PDF.

M.2 Checklist-Based MQM Pipeline

For open-ended descriptions, the evaluator receives the model description, the source figure, the expert reference, a chart-type-specific checklist, global constraints, and binding instructions. The judge first populates expected checklist values from the reference, then evaluates the model description for coverage and correctness. A rule-based engine converts checklist failures and global violations into typed penalties, deduplicates overlapping penalties, and computes the final MQM score. Binding verification checks whether labels, values, colours, and visual elements are attached to the correct entities. The full severity breakdown is reported in Appendix B.

M.3 Capability Question Pipeline

Capability questions are generated in three stages. First, GPT-4o receives the figure image, chart type, and expert description, and generates three candidates for each of four categories: counting, computation, comparison, and pattern analysis. Second, Mistral Large 3 validates candidates category-by-category against the image. Third, accepted candidates are filtered by difficulty, retaining one ques-

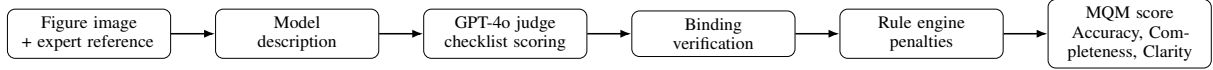


Figure 4: UML-style activity diagram for checklist-based MQM evaluation. The judge produces structured checklist judgments; deterministic post-processing maps those judgments to final scores.

Comparison	Diff	p	Sig.Stability Measure	Value
<i>Baseline MQM (vs best)</i>				
			Split-half reliability (ρ)	0.979 [0.929, 1.000]
gpt-5.2 vs gemini-3.1-pro	1.4	0.009	**Stratified ρ (Bar Chart vs Line Plot)	0.976
gpt-5.2 vs gemma3-27b-it	22.5	0.000	***Stratified ρ (Bar Chart vs Pie Chart)	0.905
gpt-5.2 vs llama4-maverick	10.2	0.000	***Stratified ρ (Line Plot vs Pie Chart)	0.952
gpt-5.2 vs phi-4-multimodal	29.5	0.000	***	
gpt-5.2 vs qwen3-vl-235b-a22b	10.7	0.000	***Scale validation (100 vs 250 figures)	
gpt-5.2 vs qwen3-vl-30b-a3b	17.1	0.000	***GPT-5.2	0.82 $\rightarrow$ 0.81
gpt-5.2 vs qwen3-vl-8b	12.7	0.000	***Gemini	0.92 $\rightarrow$ 0.91
			***Llama 4	0.78 $\rightarrow$ 0.78
			***Qwen-235B	0.74 $\rightarrow$ 0.75
<i>Resistance (vs best)</i>				
gemini-3.1-pro vs gemma3-27b-it	0.46	0.000	***Qwen-8B	0.58 $\rightarrow$ 0.57
gemini-3.1-pro vs gpt-5.2	0.10	0.000	***Qwen-30B	0.43 $\rightarrow$ 0.45
gemini-3.1-pro vs llama4-maverick	0.13	0.000	***Gemma	0.44 $\rightarrow$ 0.45
gemini-3.1-pro vs phi-4-multimodal	0.70	0.000	***Phi-4	0.22 $\rightarrow$ 0.21
gemini-3.1-pro vs qwen3-vl-235b-a22b	0.16	0.000	***	
gemini-3.1-pro vs qwen3-vl-30b-a3b	0.46	0.000	***	
gemini-3.1-pro vs qwen3-vl-8b	0.34	0.000	***	

Table 8: Paired bootstrap significance tests ($B=10,000$). Each model compared against the best. * $p < .05$, ** $p < .01$, *** $p < .001$.

Metric	GPT-4o	Mistral	Δ	p
Resistance	0.80 _{0.74} ^{0.86}	0.86 _{0.79} ^{0.91}	-0.057	0.038
Caption Bias	0.89 _{0.83} ^{0.94}	0.89 _{0.82} ^{0.94}	0.002	0.489

Table 9: Probe designer ablation. GPT-4o vs Mistral Large 3 as probe generators, GPT-5.2 as test model, GPT-4o as judge. Values show mean with 95% bootstrap CI.

tion per category and figure. Rejected candidates are written to a review queue. This produces 1,000 final questions across 250 figures.

M.4 Visual Transform and Page-Context Pipelines

We construct five transformed or page-context conditions beyond the clean image. Three are direct image transformations: Gaussian noise ($\sigma = 25$), low contrast (alpha 0.3, beta 50), and 15-degree rotation with white fill. Two use the source paper page. For the in-paper condition, the source PDF page is rendered as an image and the target figure is located on the page. For the in-paper-blur condition, the page is rendered in the same way and the target figure region is blurred. The final transformed/page-context set contains 250 noise images, 250 low-contrast images, 250 rotated images, 247 in-paper page images, and 246 in-paper-

Model	Bar	Line	Pie
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Table 11: Hallucination resistance scores by chart type.

blur images, for 1,243 cases.

M.5 Caption Bias and Resistance Pipelines

Caption bias probes test whether a model follows visual evidence or a misleading caption. GPT-4o generates a modified caption from the original caption and expert reference, embedding exactly 2–3 false but plausible claims while preserving the rest of the caption. Models then describe the figure with this modified caption. The evaluator compares each model description against the false claim and the visual reality in randomized A/B order, preventing position bias. The resistance score is the proportion of addressed claims for which the model follows the image rather than the caption.

Resistance probes use three question types per figure. *Inexist* probes presuppose a plausible but absent chart element. *Contra* probes embed a wrong numerical premise and ask the model to build on it. *Unanswerable* probes ask a domain-appropriate question that cannot be answered from the chart alone. Model responses are scored as 1.0 for clear resistance, 0.5 for partial hedging or partial compliance, and 0.0 for accepting the false premise or

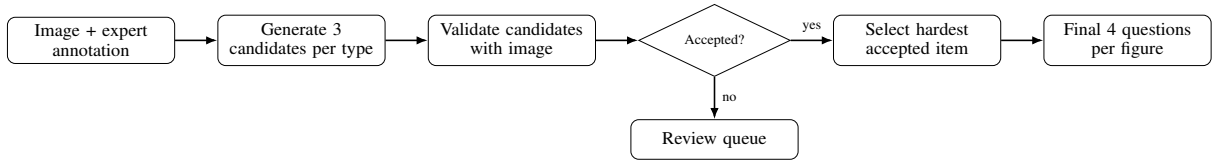


Figure 5: Capability question generation and filtering pipeline. Each figure yields one validated question for counting, computation, comparison, and pattern analysis.

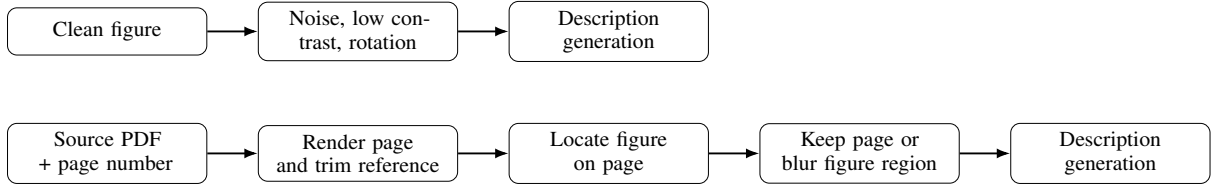


Figure 6: Pipelines for visual transformations and page-context conditions. Direct transforms alter the isolated figure; page-context conditions render the source PDF page and optionally blur the target figure region.

	MQM	Resist	Captio	Admitt	Induct
MQM	–	0.95	0.95	0.83	0.95
Resistance		–	1.00	0.86	0.93
Caption Bias			–	0.86	0.93
Admittance (Active)				–	0.88
Inductance (Active)					–

Table 12: Spearman rank correlations between evaluation dimensions ($n=8$ models). Values below 0.70 suggest the dimensions capture distinct aspects of model competence.

Model	Count	Comp.	Compar.	Pattern	Overall
Gemini	89.2	79.4	89.6	70.0	81.0
<u>GPT-5.2</u>	<u>76.1</u>	82.8	<u>77.9</u>	72.0	<u>78.4</u>
Qwen-235B	65.2	63.7	50.0	52.0	58.4
Qwen-8B	47.8	52.8	45.4	50.0	51.0
Llama 4	45.6	<u>53.4</u>	37.2	48.0	48.5
Qwen-30B	43.5	45.9	31.4	30.0	40.6
Gemma	15.2	29.4	18.6	<u>40.0</u>	27.2
Phi-4	13.0	6.2	3.5	16.0	8.0

Table 13: Capability question accuracy (%) by category. Counting, computation, comparison, and pattern analysis evaluated on 250 figures with 4 questions each. Scores averaged across two judges (GPT-4o and Mistral Large 3). **Bold** = best, underline = second best.

fabricating an answer.

M.6 Selective Blur Pipeline

Selective blur creates two visually similar but conceptually different uncertainty conditions. The pipeline first extracts OCR text and bounding boxes from each figure. GPT-4o then sees the image, OCR text list, and expert description, and proposes ranked blur targets for two cases. Admittance targets are elements whose identity should be unre-

coverable once blurred. Inductance targets are elements whose identity should remain inferable from surrounding visual context. The selected text is matched back to OCR boxes using exact, normalized, and fuzzy matching, then the target region is blended toward gray and heavily blurred. Probe files record the blurred text, target question, bounding box, and expected behaviour. Final blur candidates are manually reviewed before inclusion in the active and passive evaluations.

M.7 Probe Evaluation Rubrics

Active blur responses are judged along three binary dimensions: whether the model admits uncertainty, whether it fabricates a specific answer, and whether any fabricated answer is correct. Passive blur responses are judged similarly, but also record whether the model mentions the blurred element at all in an open-ended description. For admittance, reliable behaviour is to acknowledge unreadability and avoid unsupported specificity. For inductance, reliable behaviour is to infer correctly when the remaining visual context warrants it.

Caption bias responses are evaluated claim-by-claim. For each false claim, the judge receives the model description and two randomized statements: the modified caption claim and the visual reality. Responses are mapped back to image-following, caption-following, or not-addressed. Resistance probes are evaluated with type-specific rubrics: rejecting an absent element, correcting a false numerical premise, or stating that an unanswerable question cannot be determined from the chart.

Benchmark	Scale	Scientific figures	Open-ended perception	Capability QA	Visual stress tests	Misleading context	Selective uncertainty	Behavioural profile
ChartQA	20K+ charts, 32K+ QA	No	No	Yes	No	No	No	No
ChartBench	9K+ questions	Mixed	No	Yes	No	No	No	No
CharXiv	2K+ arXiv charts	Yes	Partial	Yes	No	No	No	No
SciFIGBench	Scientific-figure benchmark	Yes	Partial	Yes	No	No	No	No
ChartMuseum	Expert-annotated real-world chart tasks	Mixed	Partial	Yes	No	No	No	No
ChartQAPro	1,341 charts, 1,948 questions	Mixed	No	Yes	No	Partial	Partial	No
EncQA	2,076 encoding QA pairs	No	No	Yes	Partial	No	No	No
MultiChartQA	Multi-chart QA benchmark	Mixed	No	Yes	No	No	No	No
SciFIG-EVAL	250 arXiv figures, 1,000 capability questions, 1,243 transformed/page-context cases, 750 resistance probes, >23K model-output evaluations	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 14: Differentiating SciFIG-EVAL from recent chart and scientific-figure benchmarks (Masry et al., 2022; Xu et al., 2024; Wang et al., 2024; Huang et al., 2024a; Tang et al., 2025; Masry et al., 2025; Mukherjee et al., 2025; Chen et al., 2025). The comparison highlights whether each benchmark evaluates scientific figures, open-ended perception, targeted capability questions, visual stress tests, misleading context, selective uncertainty, and an explicit behavioural profile.

Probe family	Scale	Input manipulation	Target behaviour	Cognitive motivation	A-R-I axis
Visual transformations	1,243 transformed or page-context cases	Low contrast, noise, rotation, in-paper context, blurred in-paper context	Stability of perception and reasoning under degraded or contextualised visual input	Robustness under non-standard viewing conditions	Support
Caption bias	100 figures	Modified captions with plausible but incorrect claims	Whether models follow misleading context over direct visual evidence	Sycophantic agreement and anchoring	Resistance
Contradictory premises	250 figures	Questions embed a false value or relationship	Whether models reject claims contradicted by the figure	Anchoring bias	Resistance
Non-existent premises	250 figures	Questions refer to chart elements that are absent	Whether models challenge presupposed elements	Presupposition embedding	Resistance
Unanswerable premises	250 figures	Questions ask for information not present in the figure	Whether models abstain rather than fabricate	Cooperative pressure to answer	Resistance
Admittance blur	50 figures	Selective blur of unrecoverable chart elements	Whether models acknowledge visual uncertainty	Epistemic honesty under missing evidence	Admittance
Inductance blur	48 figures	Selective blur of contextually inferable elements	Whether models draw bounded inferences from partial evidence	Contextual inference under uncertainty	Inductance

Table 15: Probe taxonomy used in SciFIG-EVAL. The table summarises how each probe family maps an input intervention to a behavioural target and an A-R-I dimension.

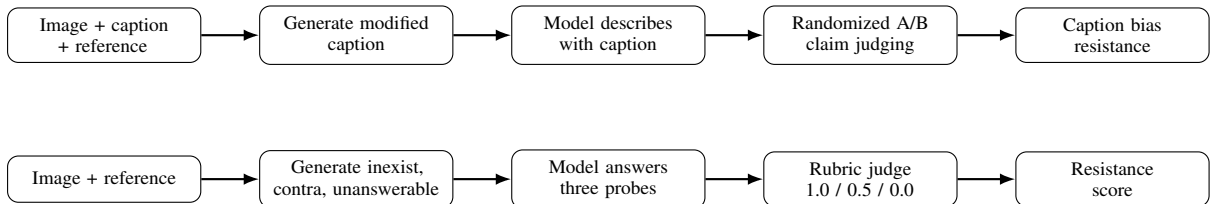


Figure 7: Pipelines for caption bias and resistance evaluation. Caption bias measures whether descriptions align with false caption claims or visual reality; resistance probes measure whether models reject false or unsupported questions.

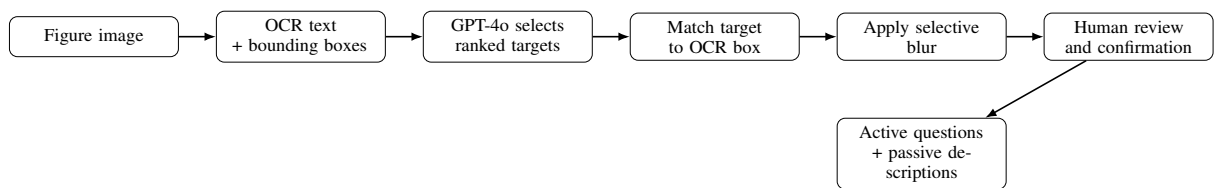


Figure 8: Selective blur pipeline for admittance and inductance probes. The same mechanism creates unrecoverable and inferable blur conditions; the distinction lies in target selection and expected behaviour.